In [1]:

```python
import boto3
import pandas as pd

s3 = boto3.client('s3')
bucket = 'cust-seg-ygp'
key = 'cleaned_online_retail_II.csv'

obj = s3.get_object(Bucket=bucket, Key=key)
df = pd.read_csv(obj['Body'])

df.head()
```

Out[1]:

| | Invoice | StockCode | Description | Quantity | InvoiceDate | Price | Customer ID | Country |
|---|---|---|---|---|---|---|---|---|
| **0** | 489434 | 85048 | 15CM CHRISTMAS GLASS BALL 20 LIGHTS | 12 | 2009-12-01 07:45:00 | 6.95 | 13085 | United Kingdom |
| **1** | 489434 | 79323P | PINK CHERRY LIGHTS | 12 | 2009-12-01 07:45:00 | 6.75 | 13085 | United Kingdom |
| **2** | 489434 | 79323W | WHITE CHERRY LIGHTS | 12 | 2009-12-01 07:45:00 | 6.75 | 13085 | United Kingdom |
| **3** | 489434 | 22041 | RECORD FRAME 7" SINGLE SIZE | 48 | 2009-12-01 07:45:00 | 2.10 | 13085 | United Kingdom |
| **4** | 489434 | 21232 | STRAWBERRY CERAMIC TRINKET BOX | 24 | 2009-12-01 07:45:00 | 1.25 | 13085 | United Kingdom |

In [8]:

```python
print(df['InvoiceDate'].dtype)
```

```
object
```

In [11]:

```python
print(df['InvoiceDate'].head())
```

```
0    2009-12-01 07:45:00
1    2009-12-01 07:45:00
2    2009-12-01 07:45:00
3    2009-12-01 07:45:00
4    2009-12-01 07:45:00
Name: InvoiceDate, dtype: object
```

In [12]:

```python
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

In [13]:

```python
print(df['InvoiceDate'].head())
```

```
0   2009-12-01 07:45:00
1   2009-12-01 07:45:00
2   2009-12-01 07:45:00
3   2009-12-01 07:45:00
4   2009-12-01 07:45:00
Name: InvoiceDate, dtype: datetime64[ns]
```

In [14]:
```python
df['Customer ID'].isna().sum()
```

Out[14]:
```
0
```

In [15]:
```python
# 'TotalPrice' to get monetary per line
df['TotalPrice'] = df['Price'] * df['Quantity']
```

In [19]:
```python
print(df['TotalPrice'].head(10))
```

```
0     83.4
1     81.0
2     81.0
3    100.8
4     30.0
5     39.6
6     30.0
7     59.5
8     30.6
9     45.0
Name: TotalPrice, dtype: float64
```

In [57]:
```python
# create a snapshot for recency
snapshot_date = df['InvoiceDate'].max() + pd.Timedelta(days=1)
print(snapshot_date)
```

```
2011-12-10 12:50:00
```

In [58]:
```python
df['RecencyTemp'] = snapshot_date - df['InvoiceDate']
```

In [59]:
```python
df.head()
```

Out[59]:

| | Invoice | StockCode | Description | Quantity | InvoiceDate | Price | Customer ID | Country | TotalPrice |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 489434 | 85048 | 15CM CHRISTMAS GLASS BALL 20 LIGHTS | 12 | 2009-12-01 07:45:00 | 6.95 | 13085 | United Kingdom | 83.4 |
| **1** | 489434 | 79323P | PINK CHERRY LIGHTS | 12 | 2009-12-01 07:45:00 | 6.75 | 13085 | United Kingdom | 81.0 |
| **2** | 489434 | 79323W | WHITE CHERRY LIGHTS | 12 | 2009-12-01 07:45:00 | 6.75 | 13085 | United Kingdom | 81.0 |
| **3** | 489434 | 22041 | RECORD FRAME 7" SINGLE SIZE | 48 | 2009-12-01 07:45:00 | 2.10 | 13085 | United Kingdom | 100.8 |
| **4** | 489434 | 21232 | STRAWBERRY CERAMIC TRINKET BOX | 24 | 2009-12-01 07:45:00 | 1.25 | 13085 | United Kingdom | 30.0 |

In [60]:
```python
# groupby
rfm = df.groupby('Customer ID').agg(
    Recency=('RecencyTemp', 'min'),
    Frequency=('Invoice', 'nunique'),
    Monetary=('TotalPrice', 'sum')
).reset_index()
```

In [61]:
```python
rfm.head()
```

Out[61]:

| | Customer ID | Recency | Frequency | Monetary |
|---|---|---|---|---|
| **0** | 12346 | 326 days 02:49:00 | 12 | 77556.46 |
| **1** | 12347 | 2 days 20:58:00 | 8 | 4921.53 |
| **2** | 12348 | 75 days 23:37:00 | 5 | 2019.40 |
| **3** | 12349 | 19 days 02:59:00 | 4 | 4428.69 |
| **4** | 12350 | 310 days 20:49:00 | 1 | 334.40 |

In [62]:
```python
# convert days
rfm['Recency'] = rfm['Recency'].dt.days
```

In [63]:
```python
rfm.head()
```

Out[63]:

| | Customer ID | Recency | Frequency | Monetary |
|---|---|---|---|---|
| **0** | 12346 | 326 | 12 | 77556.46 |
| **1** | 12347 | 2 | 8 | 4921.53 |
| **2** | 12348 | 75 | 5 | 2019.40 |
| **3** | 12349 | 19 | 4 | 4428.69 |
| **4** | 12350 | 310 | 1 | 334.40 |

In [69]:
```python
# less than 0 monetary
(rfm['Monetary'] < 0).sum()
```

Out[69]: 0

In [70]:
```python
# actual columns less than 0 monetary
rfm[rfm['Monetary'] < 0]
```

Out[70]:

| Customer ID | Recency | Frequency | Monetary |
|---|---|---|---|

In [47]:
```python
# now drop from *df*, not rfm
df = df.drop(columns='RecencyTemp')
```

In [35]:
```python
df.head()
```

Out[35]:

| | Invoice | StockCode | Description | Quantity | InvoiceDate | Price | Customer ID | Country | TotalPrice |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 489434 | 85048 | 15CM CHRISTMAS GLASS BALL 20 LIGHTS | 12 | 2009-12-01 07:45:00 | 6.95 | 13085 | United Kingdom | 83.4 |
| **1** | 489434 | 79323P | PINK CHERRY LIGHTS | 12 | 2009-12-01 07:45:00 | 6.75 | 13085 | United Kingdom | 81.0 |
| **2** | 489434 | 79323W | WHITE CHERRY LIGHTS | 12 | 2009-12-01 07:45:00 | 6.75 | 13085 | United Kingdom | 81.0 |
| **3** | 489434 | 22041 | RECORD FRAME 7" SINGLE SIZE | 48 | 2009-12-01 07:45:00 | 2.10 | 13085 | United Kingdom | 100.8 |
| **4** | 489434 | 21232 | STRAWBERRY CERAMIC TRINKET BOX | 24 | 2009-12-01 07:45:00 | 1.25 | 13085 | United Kingdom | 30.0 |

In [71]:

```
rfm.describe()
```

Out[71]:

| | Customer ID | Recency | Frequency | Monetary |
|---|---|---|---|---|
| **count** | 5878.000000 | 5878.000000 | 5878.000000 | 5878.000000 |
| **mean** | 15315.313542 | 201.331916 | 6.289384 | 2955.904095 |
| **std** | 1715.572666 | 209.338707 | 13.009406 | 14440.852688 |
| **min** | 12346.000000 | 1.000000 | 1.000000 | 2.950000 |
| **25%** | 13833.250000 | 26.000000 | 1.000000 | 342.280000 |
| **50%** | 15314.500000 | 96.000000 | 3.000000 | 867.740000 |
| **75%** | 16797.750000 | 380.000000 | 7.000000 | 2248.305000 |
| **max** | 18287.000000 | 739.000000 | 398.000000 | 580987.040000 |

In [72]:

```
import boto3
import io

# file name you want to save as
rfm_file_name = 'rfm_table.csv'

# convert RFM dataframe to CSV in memory
csv_buffer = io.StringIO()
rfm.to_csv(csv_buffer, index=False)

# upload to S3
s3 = boto3.client('s3')
bucket = 'cust-seg-ygp'    # your bucket name
```

```python
s3.put_object(
    Bucket=bucket,
    Key=rfm_file_name,
    Body=csv_buffer.getvalue()
)

print(f"Saved RFM file to s3://{bucket}/{rfm_file_name}")
```

```
Saved RFM file to s3://cust-seg-ygp/rfm_table.csv
```

In [ ]: